Human-in-the-Loop Tone Editing: Directional Semantic Optimization for Text-Guided Audio Effects

CNMAT Group 2：Song-Ze Yu, Charlotte Zhang, Yuxuan Cai

Question Statement

Abstract

- Nowadays, LLMs and generative audio models can already produce tones/music from text. *(Can Large Language Models Predict Audio Effects Parameters from Natural Language?)*

- But there is still no reliable way to **evaluate** whether those tones actually match user intention or its quality.

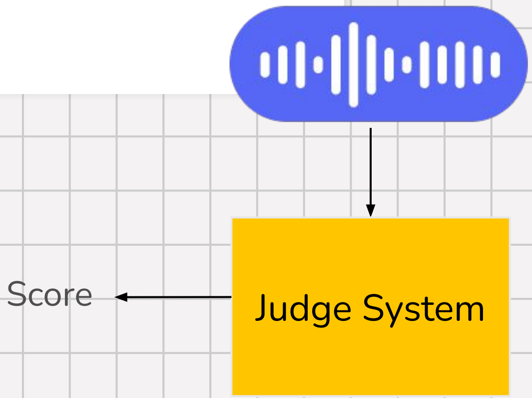
Related Works

1) **Embedding-based** evaluation (*CLAP, MERT, FAD*)
   > Only measure similarity, not quality
   > *CLAP* : general for audio to text
   > *MERT* : musical embedding to represent emotional / textual alignment

2) **Optimization-based** methods (*Text2FX*)
   > Can't evaluate output (listening test)
   > Can't refine parameters based on human feedback

Research Abstract

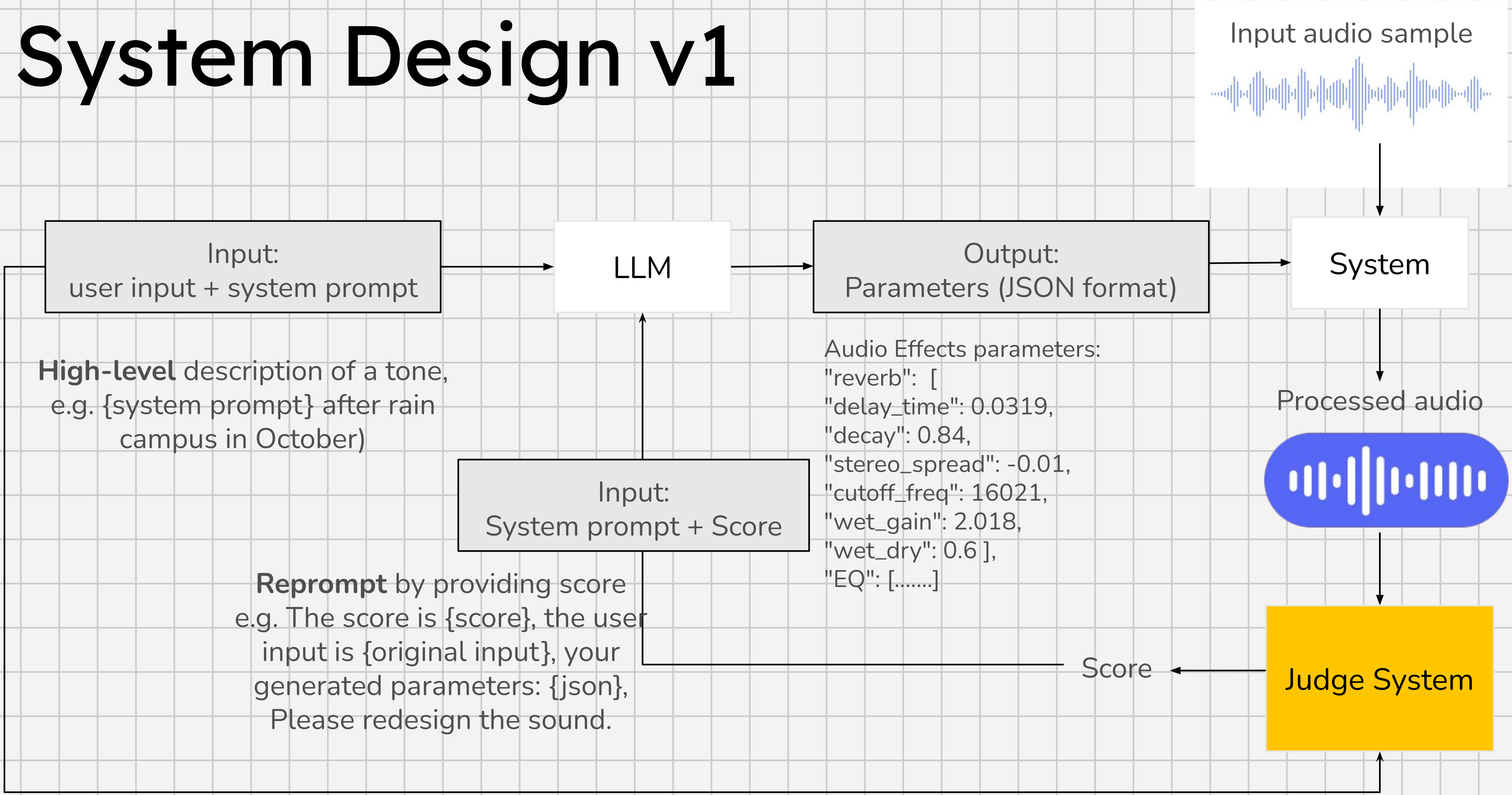
- Create an fair **evaluation system** for tone/music
- **Refine** mechanism for editing the generated tone/music (re-prompting, human-in-the-loop...)



Experiment Method v.1

A **closed-loop** architecture enabling LLMs to iteratively **refine tone design** based on **self-evaluated scores**.

System Design v1



Pipeline Demo

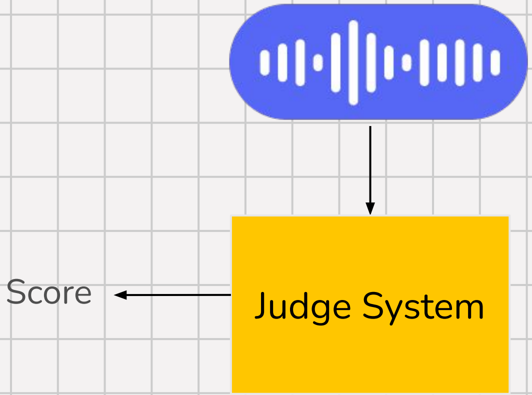
This demo will be our testing baseline for our experiment.

Dataset & Experiments (v1)

Dataset: Used **SocialFX dataset**

Method:

- Top-k CLAP similarity
- Top-k MERT + CLAP???

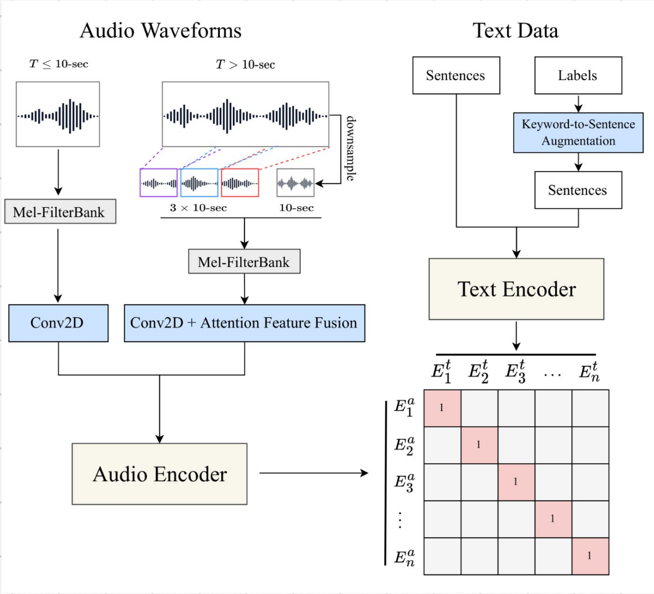


Split (3)
reverb · 6.77k rows

Search this dataset

id string · lengths 8 – 11	text string · lengths 3 – 214	param_values list · lengths 5 – 5	param_keys list · lengths 6 – 6	extra string · lengths 54 – 270
reverb_0	live-concert	[0.0319, 0.84, -0.01,…	["delay_time"…	{'lang': 'English', 'agreed': 'electric', 'didnotagree':…
reverb_1	fantasy	[0.0559, 0.94,…	["delay_time"…	{'lang': 'English', 'agreed': 'warm', 'didnotagree': 'big-…
reverb_2	dizzy	[0.0319, 0.96,…	["delay_time"…	{'lang': 'English', 'agreed': 'unpleasant,e much,echoing,distortion', 'didnotagree':…
reverb_3	echoing,chrystal,angels,heaven, harp-strums	[0.01, 0.99, -0.0043,…	["delay_time"…	{'lang': 'English', 'agreed': 'spacious', 'didnotagree': 'clashing,muted,clash,awesom
reverb_4	empty,spacious,softer	[0.0188, 0.7, 0.0035,…	["delay_time"…	{'lang': 'English', 'agreed': nan, 'didnota nan}
reverb_5	richer,exaggerated,dramatic	[0.0559, 0.94,…	["delay_time"…	{'lang': 'English', 'agreed': 'spacious', 'didnotagree': 'electric,echoes,soft,distan
reverb_6	spacious,hall,echo,vibrate	[0.0329, 0.8, 0.0044,…	["delay_time"…	{'lang': 'English', 'agreed': 'spacey,big', 'didnotagree':…
reverb_7	treble	[0.0879, 0.9, -0.0096…	["delay_time"…	{'lang': 'English', 'agreed': 'extended', 'didnotagree': 'horn-…
reverb_8	mixed,multiple-effect	[0.0109, 0.01,…	["delay_time"…	{'lang': 'English', 'agreed': 'increased-base,sharp', 'didnotagree':…
reverb_9	humongous	[0.0568, 0.91,…	["delay_time"…	{'lang': 'English', 'agreed': 'big-room,hollow,cavernous,large,big', 'didnotag
reverb_10	louder,bolder	[0.0879, 0.9, -0.0112…	["delay_time"…	{'lang': 'English', 'agreed': 'louder,loud' 'didnotagree':…
reverb_11	large,church-bells	[0.0319, 0.96,…	["delay_time"…	{'lang': 'English', 'agreed': 'melody', 'didnotagree': 'live,rock,too-…

1. Top-k CLAP

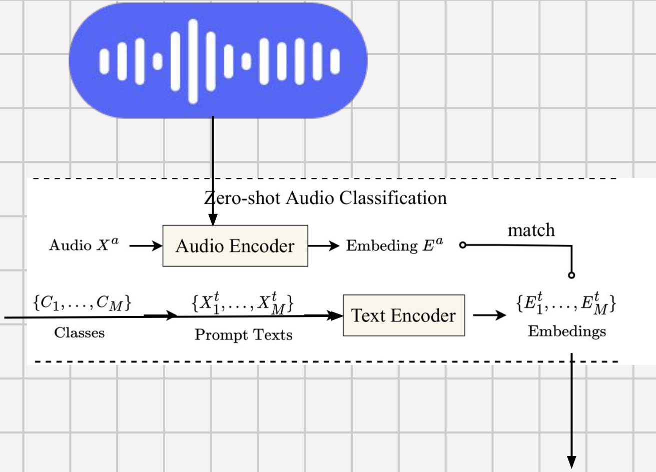


Label from SocialFX

baseline-system > data > cleaned_labels > {} reverb_labels.json

```
{
  "effect_type": "reverb",
  "labels": [
    "a-echo-howling",
    "a-nice",
    "abrasive",
    "acoustic",
    "acoustical",
    "action-invoking",
    "active",
    "affective",
    "agitated",
    "agony",
    "air-like",
    "airy",
    "alarming",
    "alert",
    "alien",
    "alive",
    "all-around",
    "alone",
    "altering",
    "alternative-rock",
    "amazing",
    "ambience",
    "ambient",
    "ambient-trails",
    "amplified",
    "ancient",
    "angelic",
    "angry",
    "annoying",
    "antique",
    "appealing",
    "approaching-stormily",
    "arena",
    "army",
    "atmospheric",
```


Processed Audio
(dry + ground truth effects)



Top-k label
(CLAP)

2. Top-k MERT

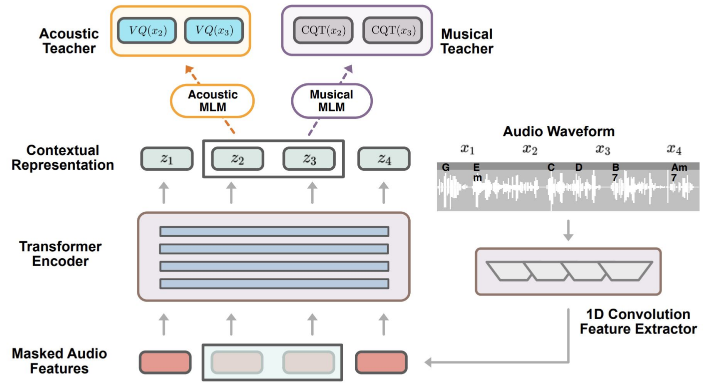
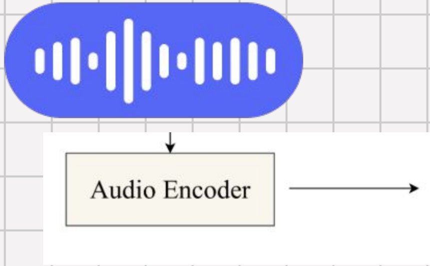


Figure 1: Illustration of the MERT Pre-training Framework.

Processed Audio
(dry + ground truth effects)



Audio embeddings
(MERT)

Linear

Audio embeddings
(CLAP)

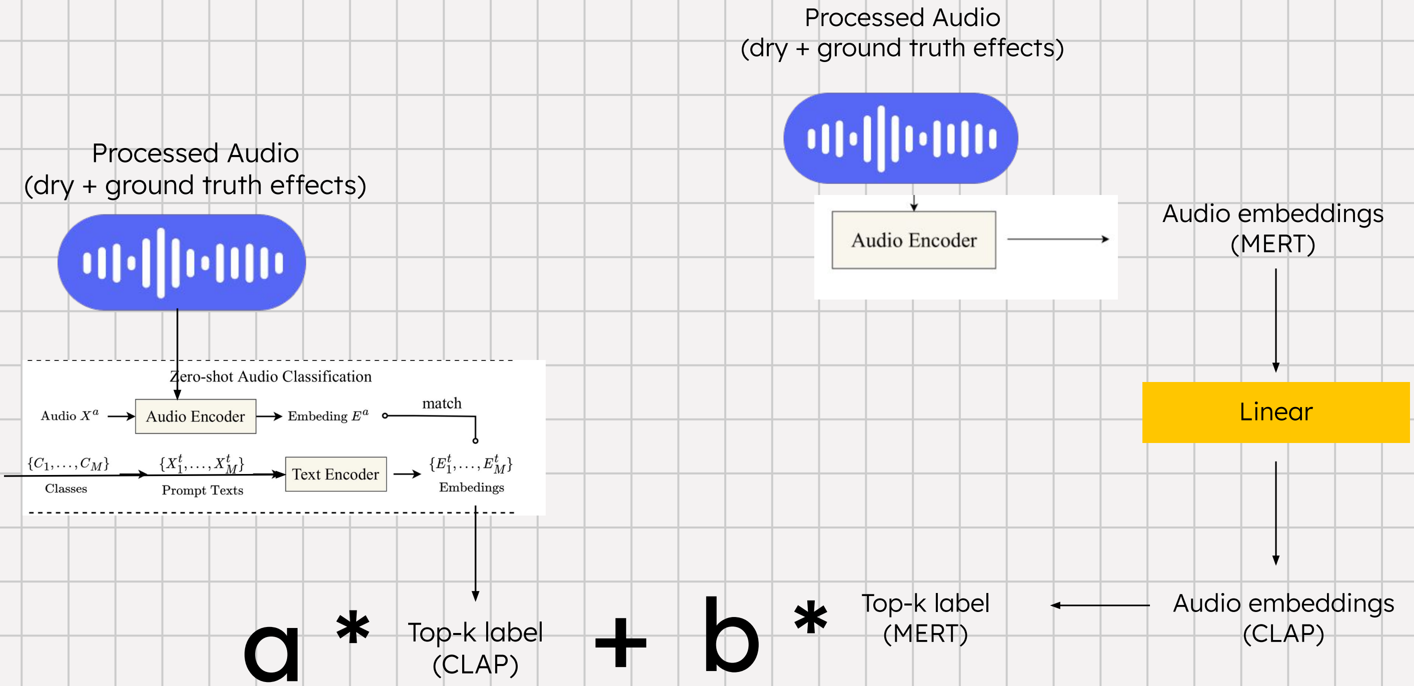
Top-k label
(MERT)

3. Hybird

Label from SocialFX

baseline-system > data > cleaned_labels > {} reverb_labels.json


```
{
  "effect_type": "reverb",
  "labels": [
    "a-echo-howling",
    "a-nice",
    "abrasive",
    "acoustic",
    "acoustical",
    "action-invoking",
    "active",
    "affective",
    "agitated",
    "agony",
    "air-like",
    "airy",
    "alarming",
    "alert",
    "alien",
    "alive",
    "all-around",
    "alone",
    "altering",
    "alternative-rock",
    "amazing",
    "ambience",
    "ambient",
    "ambient-trails",
    "amplified",
    "ancient",
    "angelic",
    "angry",
    "annoying",
    "antique",
    "appealing",
    "approaching-stormily",
    "arena",
    "army",
```

spacious.wav

```
================================================================================
SUMMARY COMPARISON
================================================================================

Rank   CLAP                          MERT                          Hybrid
------ ----------------------------- ----------------------------- -----------------------------
1      huge-trumpets                 engulfing                     classy
2      chaotic                       classy                        huge-trumpets
3      cathedral                     run-together                  antique
4      big-church                    liquid                        old-organ
5      cathedral-like                low-fidelity                  dying
6      church-bell                   roomy                         dominant
7      glorious                      wavy                          low-fidelity
8      tunnel-like                   high-bass                     airy
9      old-timely                    commanding                    serene
10     ominous                       serene                        chambers
```

warm.wav

```
================================================================================
SUMMARY COMPARISON
================================================================================

Rank   CLAP                          MERT                          Hybrid
------ ----------------------------- ----------------------------- -----------------------------
1      church-bell                   high-bass                     classy
2      big-church                    classy                        low-fidelity
3      cathedral                     low-fidelity                  high-bass
4      cathedral-like                engulfing                     chambers
5      chaotic                       roomy                         old-organ
6      church-like                   wavy                          old-timely
7      glorious                      high-pitch                    waved
8      ominous                       large-auditorium              dominant
9      chambers                      liquid                        mixed
10     classical                     comfortable                   loving
```

muffled.wav

```
================================================================================
SUMMARY COMPARISON
================================================================================

Rank   CLAP                           MERT                           Hybrid
------ ------------------------------ ------------------------------ ------------------------------
1      meditative                     high-bass                      wavy
2      lovely-echoes                  engulfing                      lovely-echoes
3      gothic                         classy                         loving
4      majestic                       wavy                           spacious
5      reflections                    robust                         soothing
6      melody                         eco                            run-together
7      introspective                  roomy                          vintage
8      jazzy                          spacious                       nicer
9      glorious                       clear                          gothic
10     mystery                        muted                          lengthy
```

No HIT in top 10......

REVERB Effect

Best Configuration: **Cleaned labels + MERT mode**

Cleaned + CLAP

837 labels | 11 samples

Hit@1:	**0.000**
Hit@5:	**0.000**
Hit@10:	**0.000**
MRR:	**0.000**

Original + CLAP

1890 labels | 11 samples

Hit@1:	**0.000**
Hit@5:	**0.000**
Hit@10:	**0.000**
MRR:	**0.000**

Cleaned + MERT

837 labels | 11 samples

Hit@1:	**0.000**
Hit@5:	**0.091**
Hit@10:	**0.091**
MRR:	**0.018**

Original + MERT

1890 labels | 11 samples

Hit@1:	**0.000**
Hit@5:	**0.000**
Hit@10:	**0.091**
MRR:	**0.013**

Cleaned + HYBRID

837 labels | 11 samples

Hit@1:	**0.000**
Hit@5:	**0.000**
Hit@10:	**0.000**
MRR:	**0.000**

Original + HYBRID

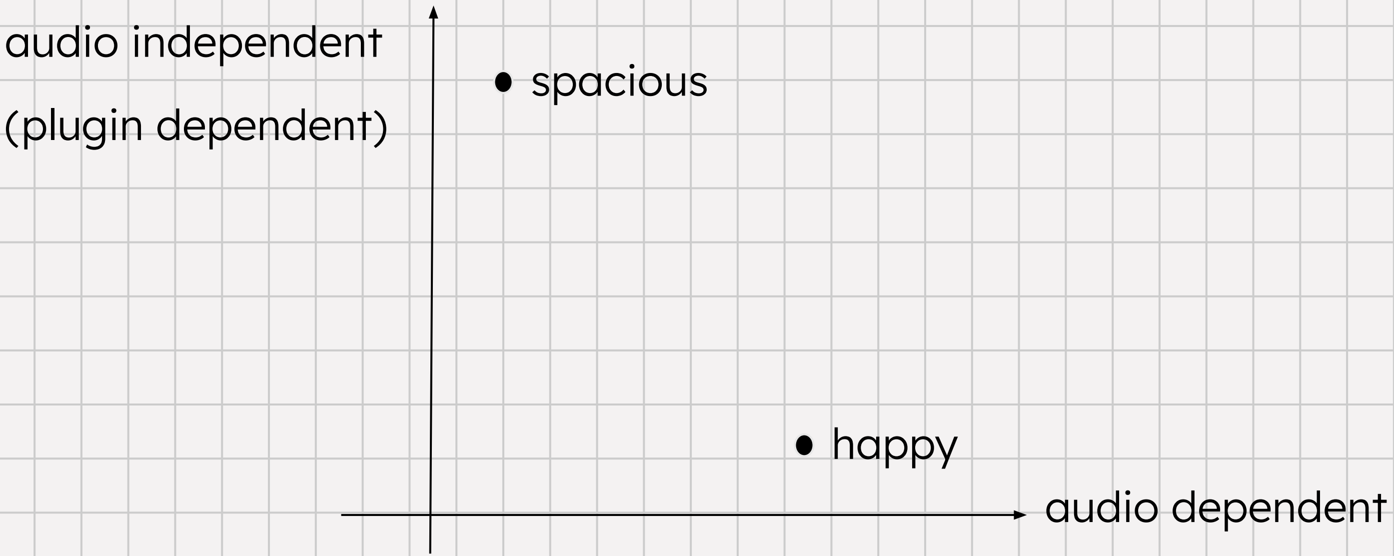
1890 labels | 11 samples

Hit@1:	**0.000**
Hit@5:	**0.000**
Hit@10:	**0.000**
MRR:	**0.000**

Hypothesis:

We assume the label can be separated into “**audio dependent**” & “audio independent”.

> **Expect plugin-dependent words to be easier ->** a better result in evaluation system.

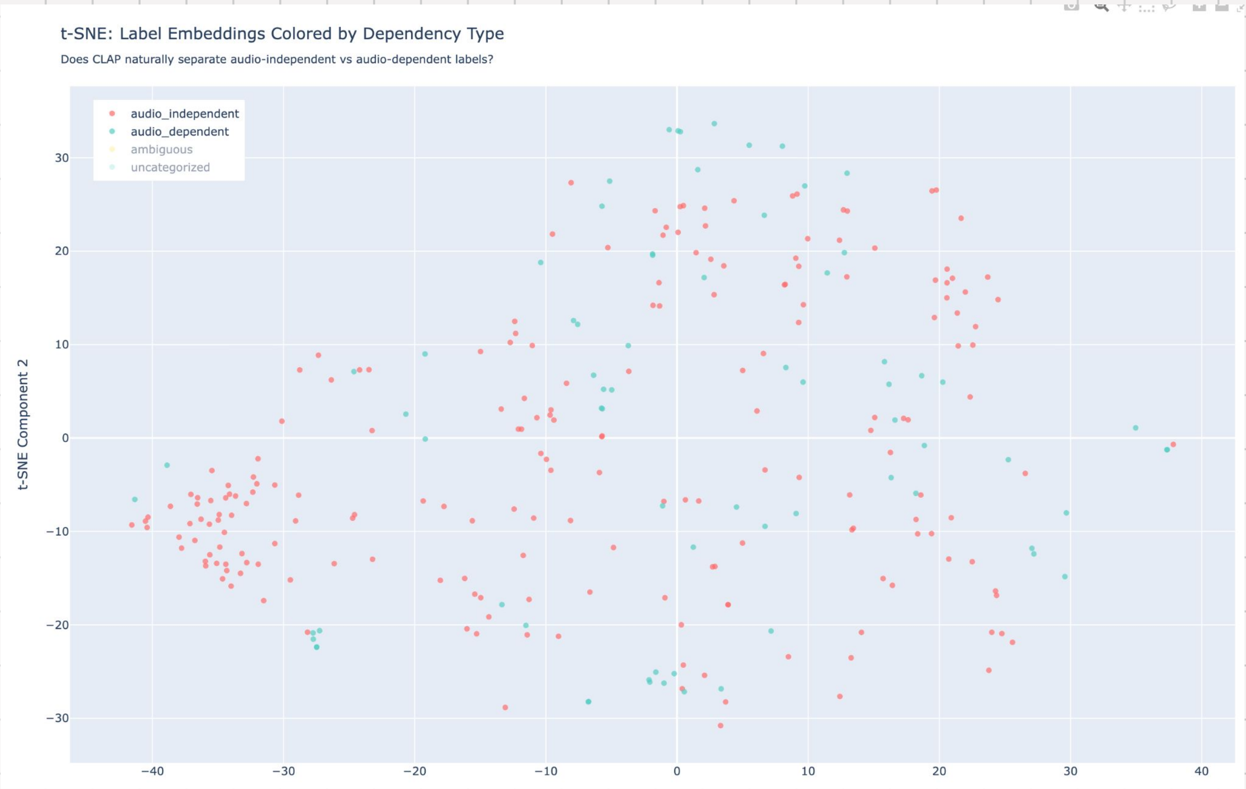


Maybe deep into the data……?

Tsne on CLAP? graphs?

It’s not separated in

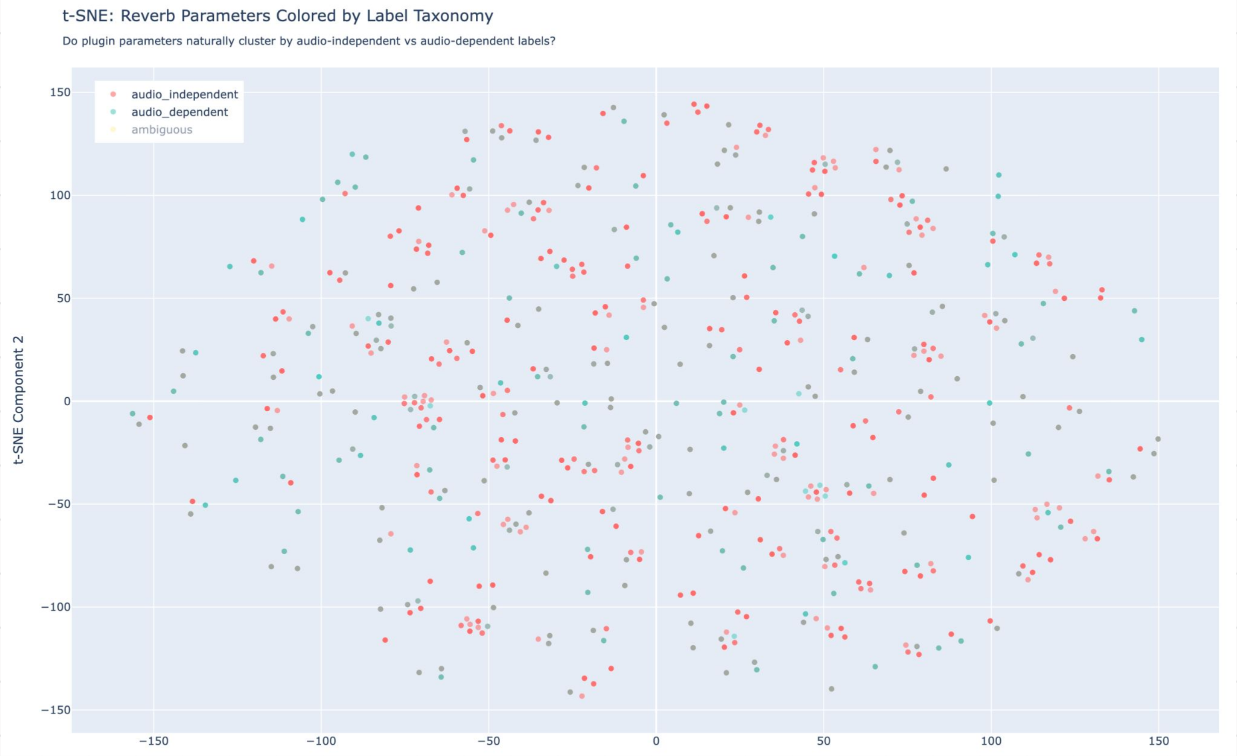
CLAP text embeddings!



Maybe deep into the data……?

Tsne on SocialFX? graphs?

descriptors are not separable!



Problem & Challenges of v1

1. It's hard to provide a fair "score" in the judge system:

- *Semantic descriptors in SocialFX are extremely noisy.*
- *No standard answer for what "warm" or "airy" should be.*
- *How to quantize music into a score?*

Human-in-the-loop!

VibeCheck!

→ Reinforces that our judge system lacks semantic grounding.

2. Reprompting may NOT converge

Even if the judge outputs a score like 3/10 (<threshold) and we reprompt the LLM:

"Given your score is 3/10, please redesign the sound."

→ LLM reprompting is fundamentally unstable.

Optimization Problem!

Question Re-Statement

Instead of asking:

"How do we ***judge*** a sound?"

We should ask:

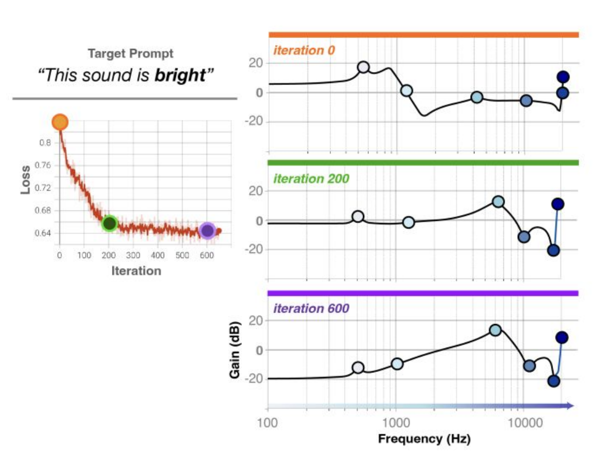
"How do we ***move*** a sound toward the user's intention?"

System Design v2: Human-in-the-Loop Optimization

Related Works

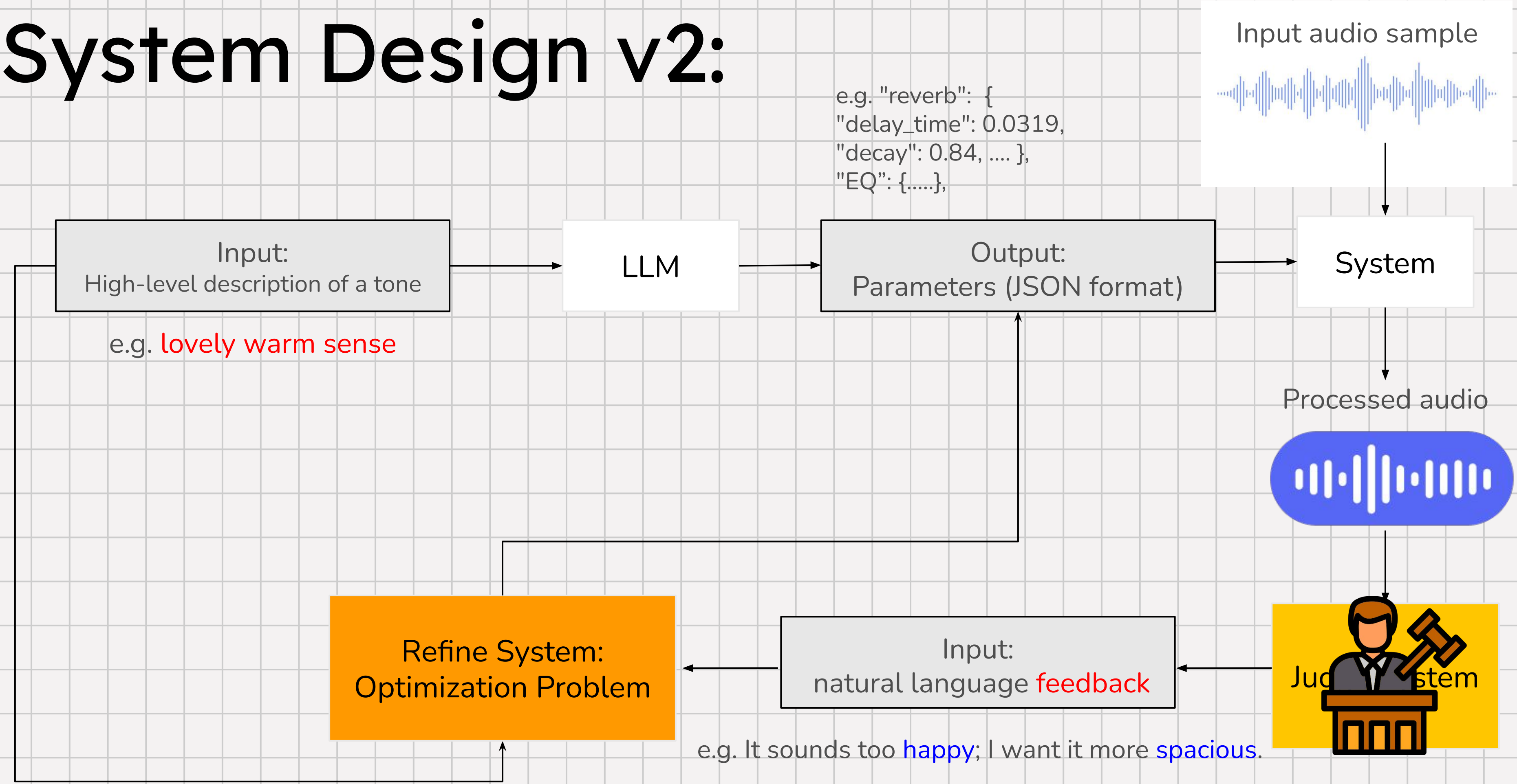
Inspired by *Text2FX*, *InstructPix2Pix*:

- *Text2FX* **optimizes** parameters toward a prompt
- InstructPix2Pix allows instruction-based refinement



We propose a **tone-editing system**, not just a tone generator.

System Design v2:



Refine System:

This avoids the need for a **hard judge** and solves the **convergence issue.**

Input:
High-level description of a tone

e.g. lovely warm sense

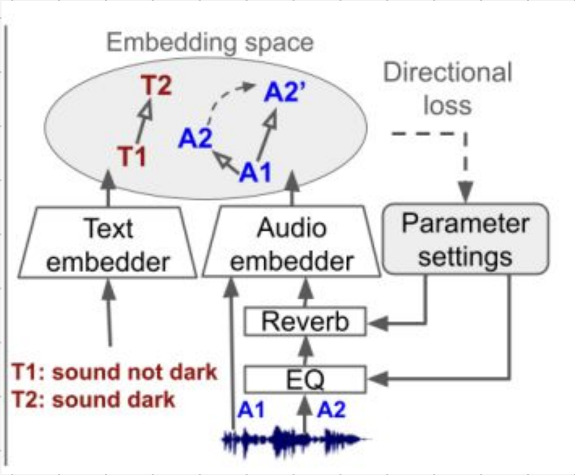
Refine System:
Optimization Problem

Input:
natural language feedback

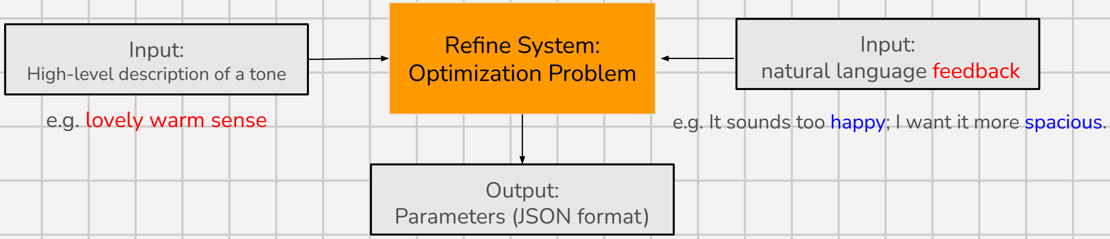
e.g. It sounds too happy; I want it more spacious.

Output:
Parameters (JSON format)

1. Compute two semantic direction vectors:
   - Original prompt direction (→ lovely, warm)
   - User-feedback direction (→ not happy, → spacious)
2. Combine directions with learnable weights
3. Apply directional update to FX parameters



Experiments:



Baseline 1 – LLM reprompt

Expected: fails (no convergence)

Baseline 2 – Feedback directional update

Optimize audio embedding movement to match feedback text movement.

Baseline 3 – Hybrid weighted direction

Combined original Prompt direction + feedback direction combined.

Baseline 4 – Gene Algorithm / Simulated Annealing

Carmine's suggestion!

Baseline 5 – RL-based refinement

Q: What **dataset** we want to use?

A: Probably still SocialFX? Or even unsupervised?

Thank you